

Workable-Items Model — every WBS task/subtask → the §11.4.93 SQLite single source of truth

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Volume 7 (Phase Execution), document 1 of 5. This spec defines the **mechanical contract** by which every Work-Breakdown-Structure leaf in the four phase WBS documents — 06-phase0-spike-wbs.md (HVPN-P0-NNN), 07-phase1-mvp-wbs.md (HVPN-P1-NNN), 08-phase2-parity-wbs.md (HVPN-P2-NNN), 09-phase3-reach-wbs.md (HVPN-P3-NNN) — becomes one row in the git-tracked SQLite single source of truth docs/workable_items.db (§11.4.93/.95). It is **spec-only**: it describes the schema, the integrity contract (§11.4.148), the per-item test diary (§11.4.149), the bidirectional docs↔DB sync via Docs Chain (§11.4.106), and the HVPN-Pn-NNN id convention (§11.4.54) — it does not build the loader (that is work-item HVPN-P1-150). Every claim here is derived from a cited phase WBS doc or constitution anchor; nothing is invented. Effort/duration figures inherited from the phase docs are sizing estimates only and are marked TARGET where a date could be inferred.

Table of contents

- 0. Why a DB-backed single source of truth
- 1. The id convention (HVPN-Pn-NNN, §11.4.54)
- 2. The canonical DDL (one table for all four phases)
- 3. Field-by-field mapping: WBS leaf → DB row

- [4. Sample rows — one per phase, drawn from the WBS docs](#)
 - [5. The §11.4.148 integrity contract \(status + type + id, everywhere\)](#)
 - [6. The §11.4.149 per-item test diary](#)
 - [7. Docs\[?\]DB sync via Docs Chain \(§11.4.106\)](#)
 - [8. The loader contract \(cmd/workable-items/, HVPN-P1-150\)](#)
 - [9. Gates, anti-bluff, and what "complete" means](#)
 - [10. Worked end-to-end example \(P0 leaf → row → diary → export\)](#)
 - [Sources verified](#)
-

0. Why a DB-backed single source of truth

The four WBS documents are human-authored Markdown — readable, reviewable, diffable. But Markdown alone is a fragile tracker: status drifts between the doc and reality, dependency cycles go undetected, and "is this work owed?" has no mechanical answer. §11.4.93 mandates a **SQLite single source of truth** for workable items, and §11.4.95 mandates that DB be **git-tracked** (it is authoritative source data, NOT a build artefact — the §11.4.77 regen-mechanism exemption explicitly does NOT apply).

The model is **bidirectional** (§11.4.93, §11.4.106): the Markdown WBS and the DB are kept byte-identical in a round-trip (md-to-db parses the leaves and upserts; db-to-md regenerates the leaf blocks). The DB is authoritative; the WBS Markdown, the rendered HTML/PDF, and any external tracker are **derived**. A verify pass is the deterministic pre-build gate that fails on any drift (§11.4.50 determinism + §11.4.106 conflict-not-silent-merge).

Each phase WBS already declares this projection in its own §3 Workable-item schema (§11.4.93 DB-ready) section: Phase 0 in 06-... §0.3, Phase 1 in 07-... §3, Phase 2 in 08-... §3, Phase 3 in 09-... §3. This document **reconciles those four into one canonical schema** and one loader contract so a single DB holds all four phases (the phase discriminator column, introduced in 08-... §3, carries P0|P1|P2|P3).

1. The id convention (HVPN-Pn-NNN, §11.4.54)

Every workable item — task or subtask — carries a stable, unique, auto-incremental ticket id (§11.4.54: append-only, never renumbered, reused, or decremented). HelixVPN uses the HVPN- project prefix instead of the generic ATM-, but the discipline is identical.

Element	Rule	Source
Prefix	HVPN- (project), then phase tag P0/P1/P2/P3	

Element	Rule	Source
		all four WBS §0
Numeric block	the epic/milestone encodes the hundreds/tens digit	07-... §0, 08-... §0, 09-... §0.1
Phase 1+ epics	Enn → tasks nn0...nn9 (E02 store → 020...029)	07-... §0
Phase 0	milestones S0...S8; gates use 001...006-style; cross-cutting 077...080	06-... §0.1
Subtask form	Phase 1+ append .k (HVPN-P1-022.3); Phase 0 allocates a fresh HVPN-P0-NNN per subtask with parent expressing hierarchy	07-... §0, 06-... §0.1
Mutability	monotonic, never renumbered ; gaps allowed; new work appends at next free number	§11.4.54, all four WBS §0
Cross-phase deps	cite the full id (HVPN-P1-090) — the namespaces are disjoint	08-... §0

Subtask-id reconciliation (R5, see REFINEMENT_NOTES.md). Phase 0 allocates a *distinct* HVPN-P0-NNN for each subtask (e.g. HVPN-P0-009/010 under task HVPN-P0-008). Phases 1–3 use the dotted .k form (HVPN-P1-022.3). Both are valid §11.4.54 stable ids; the loader (§8) treats the dotted form as a child whose parent_id is the bare task id. The three deepening docs in this volume (subtask-deepening-p1/2/3.md) author the .k subtasks that this model imports.

The HVPN-Pn-NNN[.k] string is the **binding key** across all surfaces — the DB atm_id/id primary key, the WBS heading token, and (when wired) the external tracker custom field (§11.4.148 D1). A heading-hash secondary key (§11.4.54) preserves the binding when wording reflows.

2. The canonical DDL (one table for all four phases)

The four WBS docs declare near-identical items + test_diary tables; the only divergence is column naming (id vs atm_id, est_effort_d vs effort_days) and the Phase-2 phase discriminator. The **canonical reconciliation** below is the schema the loader targets; it is a strict superset that round-trips every phase's projection without loss.

```
-- docs/workable_items.db - git-tracked (§11.4.95), single source
  of truth (§11.4.93).
-- Canonical schema reconciling 06-...§0.3 / 07-...§3 / 08-...§3 / 09-...
  §3.
```

```
CREATE TABLE IF NOT EXISTS items (
  atm_id      TEXT PRIMARY KEY,                                -- 'HVPN-
    P1-022' | 'HVPN-P0-009' | 'HVPN-P2-222.1'
  parent_id   TEXT REFERENCES items(atm_id),                  -- owning
    epic/milestone/task; NULL at epic root
  phase       TEXT NOT NULL CHECK (phase IN
    ('P0', 'P1', 'P2', 'P3')),
  kind        TEXT NOT NULL CHECK (kind IN
    ('epic', 'milestone', 'task', 'subtask', 'gate')),
  title       TEXT NOT NULL CHECK (length(title) >=
    6),
    -- DDL char-floor; loader enforces the §11.4.91 ≥6-word/
    ≥40-char rule
  description  TEXT NOT NULL CHECK (length(description) >=
    40),                -- §11.4.91/.148 D2
  status      TEXT NOT NULL DEFAULT 'Queued'
    CHECK (status IN ('Queued', 'In progress', 'Ready
    for testing',
    'In
    testing', 'Reopened', 'Operator-blocked',
    'Obsolete (→
    Fixed.md)', 'Implemented (→ Fixed.md)',
    'Completed (→ Fixed.md)', 'Fixed
    (→ Fixed.md)')),
  type        TEXT NOT NULL DEFAULT 'Task' CHECK (type IN
    ('Bug', 'Feature', 'Task')),
  severity    TEXT NOT NULL DEFAULT 'normal',
  epic       TEXT NOT NULL,
    -- 'E02' | 'S0' | 'E22'
  module      TEXT NOT NULL,                                --
    'store', 'coordinator', 'helix-core', ...
  gate       TEXT,                                           --
    'G1'..'G6', 'G20'..'G26' or NULL
  deps       TEXT NOT NULL DEFAULT '[]',                    -- JSON
    array of atm_ids (cross-phase allowed)
  deliverable TEXT NOT NULL,
  acceptance  TEXT NOT NULL,                                --
    falsifiable, captured-evidence (§11.4.5/.69/.107)
  effort_days REAL NOT NULL DEFAULT 1.0,                  -- sizing
    only - never a date (TARGET)
  test_types  TEXT NOT NULL DEFAULT '[]',                  -- JSON
    array of §11.4.169 codes
  dod_refs    TEXT NOT NULL DEFAULT '[]',                  -- JSON:
    ['AC3', 'SL01'] | ['P2-AC3'] | ['G21']
  source_refs TEXT NOT NULL DEFAULT '[]',                  -- JSON:
    ['04_P1 §7', '02 §7.2']
  created_at  TEXT NOT NULL DEFAULT (datetime('now')),
  modified_at TEXT NOT NULL DEFAULT (datetime('now')),

  -- §11.4.148 D1: status + type + id are NOT NULL by
  construction above.
```

```

-- §11.4.104: who created / who owns (canonical participant
    handles; '' = legacy, must still parse).
created_by    TEXT NOT NULL DEFAULT '',
assigned_to   TEXT NOT NULL DEFAULT ''
);

-- §11.4.34 / §11.4.93 append-only lifecycle audit (state
    transitions, NOT test runs).
CREATE TABLE IF NOT EXISTS item_history (
    id          INTEGER PRIMARY KEY AUTOINCREMENT,
    atm_id      TEXT NOT NULL REFERENCES items(atm_id),
    changed_at  TEXT NOT NULL,
    by          TEXT NOT NULL CHECK (by IN
        ('AI', 'User', 'Operator')),    -- §11.4.34
    from_status TEXT, to_status TEXT,
    reason      TEXT NOT
        NULL,                                --
        §11.4.34 closed vocab
    evidence    TEXT
        -- path/desc (§11.4.7)
);

-- §11.4.21 Operator-blocked detail (WHAT/WHY/UNBLOCK/WHO) – one
    row per blocked item.
CREATE TABLE IF NOT EXISTS operator_block_details (
    atm_id      TEXT PRIMARY KEY REFERENCES items(atm_id),
    what        TEXT NOT NULL, why TEXT NOT NULL,
    unblock     TEXT NOT
        NULL,                                --
        §11.4.148 D3 enumerated choices
    who        TEXT NOT NULL
);

-- §11.4.90 Obsolete detail (Since/Reason/Superseding/Triple-check
    evidence).
CREATE TABLE IF NOT EXISTS obsolete_details (
    atm_id      TEXT PRIMARY KEY REFERENCES items(atm_id),
    since       TEXT NOT NULL, reason TEXT NOT
        NULL,                                -- §11.4.90 closed vocab
    superseding TEXT, evidence TEXT NOT NULL
);

CREATE TABLE IF NOT EXISTS meta (
    schema_version TEXT NOT NULL,
    last_sync_at   TEXT NOT NULL,
    integrity_hash TEXT NOT
        NULL,                                -- §11.4.86
        sha256 of sorted keyset
);

CREATE INDEX IF NOT EXISTS idx_items_parent ON items(parent_id);
CREATE INDEX IF NOT EXISTS idx_items_phase  ON items(phase);

```

```
CREATE INDEX IF NOT EXISTS idx_items_gate ON items(gate);
CREATE INDEX IF NOT EXISTS idx_items_status ON items(status);
```

The gates table from 06–... §0.3 and 09–... §3 is retained verbatim as a sibling (gate verdicts are not workable items but carry their own evidence path); it is reproduced in §9.

3. Field-by-field mapping: WBS leaf → DB row

Each WBS leaf block (a task with no subtasks, or a subtask) carries the field contract declared in every phase doc's §0 Field dictionary / §0 Field contract. The mapping is 1:1:

WBS field (Markdown)	DB column	Transform / rule
heading token HVPN–Pn–NNN[.k]	atm_id	verbatim; primary key
owning epic/ milestone/task	parent_id	the immediate parent's id; epic root → NULL
Pn from the id	phase	P0/P1/P2/P3
kind (task/subtask/ epic/milestone/gate)	kind	from the heading style + §0 hierarchy
title (≥6 words, §11.4.91)	title	verbatim; loader rejects < 6 words (§11.4.91 anti-pattern)
Desc (≥40 chars, §11.4.148 D2)	description	verbatim; loader rejects stub/ section-label fragments
(lifecycle)	status	default Queued; never authored stale in the WBS
type Task Feature Bug	type	from the · type X annotation; Phase-2 default Feature, else Task
(risk)	severity	from Critical/normal annotation where present
epic id (E02/S0/E22)	epic	parsed from the id block
module: tag	module	from the epic header module: field
gate: / · DoD:	gate	G1..G6/G20..G26 or NULL
Deps	deps	JSON array; – → []; cross- phase ids allowed
Deliverable	deliverable	verbatim file paths / artefacts

WBS field (Markdown)	DB column	Transform / rule
Acceptance	acceptance	verbatim; MUST be a falsifiable captured-evidence assertion
Effort (XS(1)... XL(15) / est_effort days)	effort_days	T-shirt midpoint → REAL; sizing only, never a date (TARGET)
Tests	test_types	JSON array of §11.4.169 codes (UNIT,INT,...)
· DoD: AC3 / · SL01	dod_refs	JSON array
[04_P1 §7] inline cites	source_refs	JSON array
(attribution)	created_by/ assigned_to	§11.4.104 handles; legacy ' ' still parses

The §11.4.169 test-type code map (one canonical set + documented aliases). test_types stores canonical §11.4.169 codes. The four WBS docs accreted two shorthand dialects — Phase 0's terse abbreviations (UT/IT/CONC/CH/HQA/SC, e.g. the HVPN-P0-008 sample row and the §10 worked example) and the long forms used by Phases 1–3 (UNIT/INT/CHAOS/CHAL/SCALE/STRESS/...). The loader normalizes the Phase-0 shorthands to the canonical code on md-to-db; both render identically. Canonical codes and their §11.4.169 type (Phase-0 alias in parentheses where it differs):

Canonical code	§11.4.169 test type	P0 alias
UNIT	unit	UT
INT	integration	IT
E2E	e2e	E2E
FA	full-automation	FA
CHAL	Challenges (challenges submodule)	CH
HQA	HelixQA (test banks + autonomous sessions)	HQA
DDOS	DDoS / load-flood	DDOS
SEC	security	SEC
CHAOS	stress + chaos (chaos half)	—
STRESS	stress (load/contention/boundary)	—
CONC	concurrency / atomicity	CONC
RACE	race-condition / deadlock	—
MEM	memory (leak / soak / ceiling)	—
BENCH	benchmarking / performance	BENCH

Canonical code	§11.4.169 test type	P0 alias
SCALE	scaling	SC
UX	ux	UX
REC	recorded media evidence (§11.4.158/.159)	—
REPRO	reproducibility (bit-identical builds)	—

The only genuine alias rewrites are UT→UNIT, IT→INT, CH→CHAL, SC→SCALE; CONC, HQA, BENCH, FA, E2E, UX are already identical across phases. `validate` rejects any code outside this canonical set (or its documented alias).

Effort honesty (§11.4.6). Every phase WBS states explicitly that effort roll-ups are "indicative person-day totals ... **not** a commitment" (07–... §22, 08–... §19, 09–... §14). The loader stores `effort_days` as a sizing scalar; any report deriving a *date* from it **MUST** label the date TARGET and cite the no-commitment caveat. Phase-3 device-gated items carry the widest error bars (09–... §14) and are additionally UNCONFIRMED:/PENDING_DEVICE: (§11.4.6).

4. Sample rows — one per phase, drawn from the WBS docs

The INSERT form proves the projection is unambiguous. These rows are transcribed from the exact leaves in the WBS docs (not invented).

Phase 0 leaf — HVPN-P0-008 (plain-UDP transport baseline, 06–... §3):

```
INSERT INTO items
    (atm_id,parent_id,phase,kind,title,description,type,severity,
     epic,module,gate,deps,deliverable,acceptance,effort_days,test_types,
     dod_refs,source_refs,created_at,modified_at)
VALUES (
    'HVPN-P0-008','HVPN-P0-S0','P0','task',
    'Implement plain-UDP Transport and prove a loopback echo round-
      trip',
    'Trivial PlainUdp over tokio UdpSocket giving the throughput
      baseline every other transport's ≥50% bar (G2) is
      computed against; loopback echo validates the trait round-
      trips.',
    'Feature','normal','S0','helix-transport','G1',
    '["HVPN-P0-006"]',
    'helix-transport/src/plain_udp.rs + tests/loopback_echo.rs',
    'echo test sends 10k 1280-byte datagrams loopback, 0 lost, order-
      independent; captured CSV of round-trip count',
    0.5,'["UT","IT","CONC","BENCH"]','["G1"]','["04_P0 §4.3"]',
    '2026-06-26T12:00:00Z','2026-06-26T12:00:00Z');
```

Phase 1 leaf — HVPN-P1-022 (FORCE RLS, 07–... §7 / §3 example):

```

INSERT INTO items
    (atm_id,parent_id,phase,kind,title,description,type,severity,
    epic,module,gate,deps,deliverable,acceptance,effort_days,test_types,
    dod_refs,source_refs,created_at,modified_at)
VALUES (
    'HVPN-P1-022','HVPN-P1-E02','P1','task',
    'FORCE RLS + tenant_isolation policies on all 10 tenant tables',
    'Enable + FORCE row-level security with a tenant_isolation USING/
    WITH CHECK policy on every tenant-scoped table and abort
    startup if the app role can bypass RLS.',
    'Task','high','E02','store',NULL,
    '["HVPN-P1-021"]',
    'migrations/0002_rls.sql; store/rls_guard.go startup check',
    'a crafted cross-tenant read returns ZERO rows (captured);
    startup aborts when role has rolsuper/rolbypassrls; CHAOS
    drop mid-tx rolls back, no leak.',
    4, '["UNIT","INT","SEC","CHAOS"]', '["AC8","AC2","AC3"]', '["02
    $2.3","04_P1 $2.2"]',
    '2026-06-26T12:00:00Z','2026-06-26T12:00:00Z');

```

Phase 2 leaf — HVPN-P2-222 (UDP hole punching, 08–... §8 / §3 example):

```

INSERT INTO items
    (atm_id,parent_id,phase,kind,title,description,type,severity,
    epic,module,gate,deps,deliverable,acceptance,effort_days,test_types,
    dod_refs,source_refs,created_at,modified_at)
VALUES (
    'HVPN-P2-222','HVPN-P2-E22','P2','task',
    'UDP hole punching across the NAT-type matrix',
    'Drive simultaneous WG handshake probes to every peer candidate
    so full-/restricted-/port-restricted-cone NATs open a
    direct path, latching WireGuard roaming onto the working
    endpoint; symmetric pairs fall through to relay.',
    'Feature','high','E22','helix-core',NULL,
    '["HVPN-P2-220","HVPN-P2-221"]',
    'helix-core/src/nat/punch.rs; netns NAT-matrix harness',
    'direct datagrams captured bypassing the gateway for full/
    restricted/port-restricted pairs; symmetric pairs cleanly
    fall through to relay; no false direct claim
    (§11.4.107).',
    8, '["UNIT","E2E","STRESS","SEC","CHAL"]', '["P2-AC3","P2-
    SL01"]', '["04_P2 §3.2","SYNTHESIS §3 D6"]',
    '2026-06-26T12:00:00Z','2026-06-26T12:00:00Z');

```

Phase 3 leaf — HVPN-P3-211 (HarmonyOS VPN ability, 09–... §6 / §3 seed):

```

INSERT INTO items
    (atm_id,parent_id,phase,kind,title,description,type,severity,
    epic,module,gate,deps,deliverable,acceptance,effort_days,test_types,
    dod_refs,source_refs,created_at,modified_at)
VALUES (
    'HVPN-P3-211','HVPN-P3-E21','P3','task',

```

```
'HarmonyOS Network Kit VpnExtensionAbility + lifecycle + kill-
switch wiring',

    'ArkTS VpnExtensionAbility opens the Network Kit tunnel
fd, hands it to the helix-core .so over NAPI, and drives
connect/disconnect/kill-switch from the core status
stream; proven by tunnel UP carrying traffic on a real
device.',
'Feature', 'Critical', 'E21', 'shims/harmonyos', 'G21',
['HVPN-P3-210', 'HVPN-P3-201'],
'shims/harmonyos/entry/src/main/ets/HelixVpnAbility.ets + napi
bridge',
'PENDING_DEVICE: on a HarmonyOS NEXT device, enroll->UP->curl
reaches authorized LAN host; window-scoped MP4 + DevEco
heap capture; §11.4.3 SKIP hardware_not_present until G20
provisions a device.',
8.0, ['UNIT', 'INT', 'E2E', 'MEM', 'UX', 'REC', 'CHAL'], ['G21'], ['04_UI
§6.1', '04_ARCH §5.7'],
'2026-06-26T12:00:00Z', '2026-06-26T12:00:00Z');
```

5. The §11.4.148 integrity contract (status + type + id, everywhere)

§11.4.148 binds the workable-item discipline into one DB[?]docs[?]tracker integrity contract. The five clauses, instantiated for HelixVPN:

- **D1 — no item without a valid status + type + id, on ALL surfaces.**
The DDL enforces atm_id (PK, NOT NULL), status (NOT NULL, closed CHECK set), and type (NOT NULL, closed CHECK set). The loader rejects a WBS leaf missing any of the three; the db-to-md regen always emits all three; the external tracker push (D5) carries all three. The id is the cross-surface binding key.
- **D2 — comprehensive structured description.** description ≥ 40 chars (§11.4.91) carrying WHAT + HOW-it-manifests + acceptance intent. The *reproduce* and *acceptance-criteria* facets live in acceptance (a falsifiable captured-evidence assertion per §11.4.5/.69/.107). The loader rejects §11.4.91 anti-pattern fragments (Composes with, bare Critical, a §-letter alone).
- **D3 — BLOCKED items carry WHY + enumerated unblock CHOICES.**
Any Operator-blocked item (§11.4.21) MUST have an operator_block_details row enumerating the closed list of unblock choices ([A]... [B]... [C]..., §11.4.66 shape). Phase 3 is the heaviest user: HVPN-P3-211/221 carry unblock = "provision a HarmonyOS NEXT / Aurora device + add to the §11.4.128 tracked-device set"; HVPN-P3-252 (audit engagement) carries unblock = "fund + contract an independent audit firm (§11.4.101 spend block)".

- **D4 — never-missed bidirectional DB↔docs↔tracker sync** — §7 below.
- **D5 — generic idempotent external-tracker push** — statuses (collapsed onto the tracker's native set per §11.4.33/.112, precise value preserved in a header), types, assignee (§11.4.104 handle from a project env var, never hardcoded/logged §11.4.10), and per-item sub-tasks; match-by-stable-key [HVPN-Pn-NNN], present⇒UPDATE/absent⇒CREATE, sink-side created=N updated=M failed=0 proof (§11.4.69). The tracker, list/board id, and field map are consumer-registered at runtime (§11.4.28 decoupling), never hardcoded.

A release tag is blocked while any item violates D1–D3 (07–... §3 anti-bluff note; §11.4.148 release-blocker clause).

6. The §11.4.149 per-item test diary

§11.4.149 mandates an append-only **testing diary** — one row per test *run* against an item — distinct from `item_history` (lifecycle state transitions). The schema (already declared in 07–... §3 and 08–... §3, restated canonically):

```
CREATE TABLE IF NOT EXISTS test_diary (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  atm_id      TEXT NOT NULL REFERENCES items(atm_id),
  date_time   TEXT NOT NULL,
              -- ISO-8601 UTC
  tested_by   TEXT NOT NULL CHECK (tested_by IN
    ('User', 'Operator', 'AI-agent', 'HelixQA')),
  result      TEXT NOT NULL CHECK (result IN
    ('PASS', 'FAIL', 'SKIP')), -- §11.4.45 vocab
  test_type   TEXT NOT NULL,
              -- one §11.4.169 code
  evidence_path TEXT,
              -- qa-results/<run-id>/...
  feature_class TEXT,
              -- §11.4.69 sink-side taxonomy class
  action_taken TEXT NOT NULL DEFAULT '',
  status_changed TEXT NOT NULL DEFAULT '',
              -- from→to if this run flipped status
  observations TEXT NOT NULL,
              -- in-depth, facts or UNCONFIRMED: (§11.4.6)
  -- THE anti-bluff constraint: a PASS row is impossible without
  -- captured evidence.
  CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND
    length(evidence_path) > 0))
);
```

The CHECK (result <> 'PASS' OR evidence_path ...) clause makes a PASS-bluff *mechanically impossible at the schema layer*: SQLite itself rejects a PASS row with no evidence path (07-... §3, §11.4.149(a)). The diary feeds §11.4.132 risk-ordered validation (most-recently-tested / most-reopened first) and the per-item Diary/Diary_Summary four-format exports (§11.4.65, §11.4.149(c)).

Tooling is minimal-LLM and deterministic (§11.4.149(e)): the observations prose is authored by whoever ran the test; the tooling only stores / renders / pushes / validates. The external-tracker SUB-TASK lifecycle (TODO|In-progress|Completed, §11.4.149(d)) reuses the D5 idempotent push.

7. Docs[?]DB sync via Docs Chain (§11.4.106)

The WBS Markdown $\rightleftharpoons$ DB $\rightleftharpoons$ exports round-trip is mechanized by **Docs Chain** (vasic-digital/docs_chain, §11.4.106) — the canonical bidirectional document-and-database propagation engine, consumed by **reference** (never copied) and configured as data via .docs_chain/contexts/wbs.yaml (07-... §20 HVPN-P1-150, 08-... §16 HVPN-P2-300, 09-... §12 HVPN-P3-272).

```
# .docs_chain/contexts/wbs.yaml (consumer-owned, §11.4.28
#   decoupling)
context: helixvpn-wbs
sources:
  - path: docs/research/mvp/final/06-phase0-spike-wbs.md      #
    #   leaf blocks → items(phase=P0)
  - path: docs/research/mvp/final/07-phase1-mvp-wbs.md        #
    #   → items(phase=P1)
  - path: docs/research/mvp/final/08-phase2-parity-wbs.md      #
    #   → items(phase=P2)
  - path: docs/research/mvp/final/09-phase3-reach-wbs.md       #
    #   → items(phase=P3)
  - path: docs/research/mvp/final/v07-execution/subtask-deepening-
    p1.md # .k subtasks
  - path: docs/research/mvp/final/v07-execution/subtask-deepening-
    p2.md
  - path: docs/research/mvp/final/v07-execution/subtask-deepening-
    p3.md
db: docs/workable_items.db                                     #
#   git-tracked (§11.4.95)
exports: [html, pdf]                                          #
#   §11.4.65 siblings; DOCX added for Status set (§11.4.153)
change_detection: content-hash                                #
#   §11.4.86 (NOT mtime)
on_conflict: surface                                          #
#   §11.4.6 – never silent-merge
evidence_dir: qa-results/docs_chain/<run-id>/                 #
#   §11.4.69 captured per run
```

Docs Chain guarantees (§11.4.106): content-hash change detection (§11.4.86, not mtime), atomic-rename + SQLite-txn commit + rollback (§9.2), both-dirty sync → conflict-surfaced-not-merged (§11.4.6), verify as the deterministic pre-build gate (§11.4.50), per-run captured evidence (§11.4.69), and an honest typed ToolAbsentError + §11.4.3 SKIP when pandoc/weasyprint is missing (never a fake transform). The meta.integrity_hash (sha256 of the sorted item keyset, §11.4.86) is the drift-proof fingerprint the freshness gate compares.

8. The loader contract (cmd/workable-items/, HVPN-P1-150)

A deterministic Go binary (NOT LLM-driven in the data path, §11.4.149(e)) implements the projection. It lives in the constitution submodule for cross-project reuse (§11.4.74, §11.4.93). Subcommands:

Subcommand	Behaviour
md-to-db	parse every WBS leaf block → upsert items/item_history by atm_id; reject §11.4.91/.148-D1/D2 violations
db-to-md	regenerate the leaf blocks from the DB — byte-identical round-trip (§11.4.93, closed-set whitespace/section-order tolerance)
diff	show md↔db divergence; non-empty diff blocks commit (commit_all.sh refuses, §11.4.93)
validate	run in the pre-build sweep — enforce id uniqueness/monotonicity (§11.4.54), dependency-DAG acyclicity, status/type closed sets, the §11.4.149 PASS-needs-evidence constraint
add / close	structured mutation that stages+commits+pushes the DB alongside the MD regen (§11.4.95, §2.1 multi-upstream) with a WAL checkpoint before stage

The loader is itself 100% test-covered (§11.4.27): unit (parse/regen), integration (against a real SQLite + a real external tracker with an honest §11.4.3 SKIP when the token is absent), the byte-identical round-trip property test (§11.4.93), export, a HelixQA Challenge (CME-WORKABLE-ITEMS-001-class), and a paired §1.1 mutation (corrupt a row → validate FAILs).

9. Gates, anti-bluff, and what "complete" means

The gates sibling table (verbatim from 06–... §0.3 + 09–... §3) tracks the phase exit gates that unblock/close work, with their own evidence path:

```
CREATE TABLE IF NOT EXISTS gates (
  id          TEXT PRIMARY KEY,          --
            'G1'..'G6','G20'..'G26'
  question    TEXT NOT NULL,
  go_no_go_bar TEXT NOT NULL,
  section      TEXT NOT NULL,
  owning_epic  TEXT,
  outcome      TEXT NOT NULL DEFAULT 'pending'
            CHECK (outcome IN
            ('pending','pass','fail','rust','go',
            'pending_device','pending_toolchain','operator_blocked')),
  evidence_path TEXT
);
```

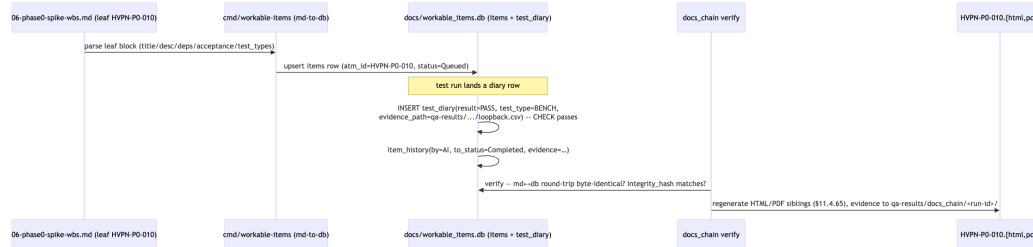
A leaf is complete only when (07–... §0 anti-bluff note, §11.4.93/.149):

1. its required §11.4.169 test_types are all green **with** a test_diary row per type carrying a non-empty evidence_path (the schema CHECK enforces this for PASS rows);
2. where it has a user-visible surface, a window-scoped MP4 (§11.4.154/.155) vision-verified through the §11.4.163 media-validation pipeline exists;
3. its lifecycle reached a terminal status (Fixed/Implemented/Completed (→ Fixed.md)) with an item_history closure row (§11.4.34 By/Reason/Evidence);
4. its gate (if any) has outcome IN ('pass','rust','go') — or, for Phase-3 device-/engagement-/toolchain-gated items, an honest pending_device/pending_toolchain/operator_blocked (e.g. G26 reproducibility held PENDING_TOOLCHAIN: per 09–... R-P3-7) with the §11.4.148 D3 unblock condition, **never** a faked pass (§11.4.6, 09–... §13 honest-gap rule).

Metadata-only / config-only / grep-without-runtime PASS is forbidden (§11.4 / §11.4.1). The §11.4.147 agent registry treats any non-complete item as work still owed; the §11.4.126 endless-loop done-condition cannot read satisfied while any item is non-terminal.

10. Worked end-to-end example (P0 leaf → row → diary → export)

Tracing HVPN-P0-010 (loopback echo soak, 06-... §3) through the whole model:



1. The leaf in 06-... §3 declares acceptance: 0 lost @ loopback; deterministic over N=3 (§11.4.50) and test_types: [IT, CONC, BENCH, FA].
2. md-to-db upserts the items row with status='Queued', phase='P0', parent_id='HVPN-P0-008'.
3. A real run captures qa-results/<run-id>/loopback_echo.csv; a test_diary row records result='PASS', test_type='BENCH', evidence_path=... — the schema CHECK admits it *only because* the evidence path is non-empty.
4. The N=3 determinism (§11.4.50) lands three identical-hash diary rows; an item_history row flips status → 'Completed (→ Fixed.md)'.
5. docs_chain verify confirms the md↔db round-trip is byte-identical and the meta.integrity_hash matches, then regenerates the HTML/PDF siblings.

The same flow applies to every leaf across all four phases; the three deepening docs in this volume author the .k subtask leaves that feed it.

Sources verified

- 06-phase0-spike-wbs.md (Phase 0 WBS, §0.3 DDL, §0.4 test-type map, gates table) — read 2026-06-26.
- 07-phase1-mvp-wbs.md (§3 items/test_diary DDL + sample row, §0 field contract, §20 HVPN-P1-150 loader, §22 effort caveat) — read 2026-06-26.
- 08-phase2-parity-wbs.md (§3 DDL with phase discriminator + sample row, §16 HVPN-P2-300) — read 2026-06-26.
- 09-phase3-reach-wbs.md (§3 DDL + seed row, §12 HVPN-P3-272, §13 honest-gap rule, §14 risk register) — read 2026-06-26.
- REFINEMENT_NOTES.md R5 (subtask-tier asymmetry / DB import driver) — read 2026-06-26.

- Constitution CLAUDE.md anchors

§11.4.34/.54/.65/.74/.86/.90/.91/.93/.95/.104/.106/.132/.147/.148/.149/.169 — project context, read 2026-06-26.

Honest boundary (§11.4.6): this document specifies the model and the loader contract; it does **not** assert the loader is implemented (that is the open work-item HVPN-P1-150). All effort/duration figures are sizing-only TARGETs per the phase docs' no-commitment caveat.